HUANG HUA-YU (黃樺裕)

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EDUCATION

TAMKANG UNIVERSITY — B.B.A. in Information Management; GPA: 3.95/4.00 (Top 1%) Sep 2021 – Jun 2025

Coursework: Data Structures, Algorithms, Machine Learning, Deep Learning, Linear Algebra, Statistics, Discrete Mathematics, Operating Systems, Database Systems, System Analysis & Design, Information Security, Cloud Service Architecture (AWS)

SKILLS

Programming: Python (NumPy, Pandas, Scikit-learn, TensorFlow, OpenCV, Flask), C/C++, JavaScript (Vue.js, Vite), SQL, C# (ASP.NET)

Machine Learning: Regression, Ensemble Methods (Random Forest, XGBoost, LightGBM), Feature Engineering, PCA/LDA, Cross-Validation & Hyperparameter Tuning, Deep Learning (DNN, CNN), Computer Vision (YOLOv8), NLP Preprocessing (TF-IDF, Tokenization, Stemming)

Tools & Infrastructure: Linux, Docker, Git/GitHub, CI/CD, MySQL, PostgreSQL, Oracle, TensorBoard, Power BI, AWS (Cloud Architecture)

Languages: Chinese (Native), English

PROJECTS

Wearable Navigation Device for the Visually Impaired — Capstone Project (Department Award Winner)

- Built a real-time obstacle detection system combining **YOLOv8** image recognition, smartphone camera, voice alerts, and ultrasonic sensors to improve safety and independence for visually impaired users.
- Owned the vision-module integration and embedded sensor pipeline (ESP8266, C++), applying low-pass filtering and Wi-Fi web display for distance feedback.
- Achieved **95% obstacle detection accuracy** with precise distance reporting to the mobile client; won the departmental project award.

ML Model Benchmark & Feature Engineering Pipeline — Python, Scikit-learn (github.com/cdexswzaq0110/ML)

- Benchmarked **6 classification algorithms** across breast cancer, IMDb, and multiple public datasets; applied PCA, LDA, and Random-Forest-based feature selection and dimensionality reduction.
- Implemented NLP preprocessing (TF-IDF, tokenization, stemming, stop-word removal) and systematic hyperparameter tuning with cross-validation, quantifying the accuracy gains from feature engineering on each dataset.

House Price Prediction with Deep Learning — TensorFlow

- Designed and trained a 3-layer fully connected neural network on the King County housing dataset; handled preprocessing, feature standardization, and inverse-transform prediction.
- Integrated **TensorBoard** monitoring with automatic best-model checkpointing; reached **~13% MAPE** on the validation set across multiple model configurations.

Data Structures & Algorithms Library — C++ (github.com/cdexswzaq0110/DSA)

- Maintained a public C++ repository of trees, graphs, dynamic programming, and string-matching implementations (CLRS-style) for competitive programming and LeetCode practice.

OrdoStack — Local-First Daily Planning & MLOps System — Python, Docker, MySQL, ClearML, AWS

- Built a local-first daily planning and MLOps system with task scheduling, execution logs, duration prediction, daily analytics, MySQL persistence, Docker Compose, ClearML tracking, backup/restore, and AWS-ready deployment configuration.

Titanic ML Control Tower — RESTful API + Machine Learning Dashboard — Flask, Python, Scikit-learn

- Extended a Titanic RESTful API and CRUD system into an ML control dashboard with dataset overview, model training, version management, single prediction, What-if simulation, CSV batch prediction, and feature importance.
- Implemented one-click model training for Logistic Regression, Random Forest, Gradient Boosting, Extra Trees, and SVM with GridSearchCV search modes and Ajax-style training status polling.
- Tracked model performance using Accuracy, Precision, Recall, F1, ROC-AUC, Confusion Matrix, joblib pipelines, JSON metadata, Active Model, and Champion Model selection.

System Design Portfolio — Local-First API Prototypes — Python, FastAPI, SQLite, Testing

- Built FastAPI prototypes for QR short links, Markdown Q&A retrieval, and task scheduling, with SQLite persistence, validation tests, API docs, runbooks, and browser UIs.

ACTIVITIES & CERTIFICATIONS

- Python AI Application Engineer Training Program — Institute for Information Industry (III)
- Getting Started with AI on Jetson Nano — NVIDIA; Introduction to Cybersecurity — Cisco
- 2025 National Programming Contest; 2024 Innovative ICT Application Contest; 2023 Golden Shield Information Security Competition; CYBERSEC 2023 Taiwan